The Cycle Class of a Subvariety and the Reverse Arrow of the Hodge Conjecture

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This note typesets, without omission, the author's written account of the cycle class map and of what remains open in the Hodge conjecture. It does not add a construction of the inverse. The Clay statement is rational. The integral Hodge conjecture is stated separately below and is not claimed as a theorem of this note.

The last geometric object on screen is not a mystery class. It is the cycle class of a subvariety: the one arrow that is already a theorem. The video's closing line, "Decoding still required," refers to the reverse arrow, which remains open.

What the final object is

In Step 05 the film names a closed algebraic subset

$$Z \subset X$$

of complex codimension k, and then writes its class

$$[Z] \in H^{2k}(X, \mathbb{Q}) \cap H^{k,k}(X).$$

That symbol $[Z]$ is the cycle class of Z. It is the source of the "balanced signal." Every later frame — the conjecture, the scoreboard, and the card "Hodge classes $\to$ algebraic cycles" — is asking whether an arbitrary Hodge class is a rational combination of objects of this same kind.

The yellow loop drawn on the wireframe is a picture, not a construction. A genuine algebraic subvariety of complex codimension k is cut out by polynomial equations and has complex dimension $\dim X - k$. On an elliptic curve the algebraic classes in positive codimension are points, not real circles. The animation is showing the role of Z, not a particular equation.

How the class is produced

Let X be a smooth complex projective variety of dimension n, and let $Z \subset X$ be a closed irreducible subvariety of codimension k. The complex structure orients the regular locus of

Z as a real $2(n-k)$-dimensional manifold. That orientation supplies a fundamental class in homology,

$$[Z]_{\mathrm{hom}} \in H_{2n-2k}(X, \mathbb{Z}).$$

Poincaré duality on the compact oriented manifold X converts it into a cohomology class

$$[Z] \;:=\; \mathrm{PD}\big([Z]_{\mathrm{hom}}\big) \in H^{2k}(X, \mathbb{Z}) \subset H^{2k}(X, \mathbb{Q}).$$

Equivalently, Z defines a closed current of integration: for a smooth form φ of degree $2n-2k$,

$$T_Z(\varphi) \;:=\; \int_{Z^{\mathrm{reg}}} \varphi.$$

The current T_Z is closed, of type (k,k), and represents the same class. Extending by linearity from irreducible subvarieties to rational algebraic cycles gives the cycle class map

$$\mathrm{cl}\colon \quad Z^k(X)_{\mathbb{Q}} \;\longrightarrow\; H^{2k}(X, \mathbb{Q}).$$

The image of cl is what the video calls the algebraic classes. The integral cycle-class map, with values in $H^{2k}(X, \mathbb{Z})$, is the same construction before tensoring with $\mathbb{Q}$.

Why the class is automatically a Hodge class

A current of type (k,k) pairs to zero against every form of type (p,q) with $(p,q) \neq (n-k, n-k)$. Passing to cohomology, the class therefore has vanishing off-diagonal Hodge components:

$$[Z] \in H^{k,k}(X).$$

Combined with rationality,

$$[Z] \in H^{2k}(X, \mathbb{Q}) \cap H^{k,k}(X).$$

That intersection is the definition of a Hodge class. This is the easy arrow on screen:

$$\text{geometry } Z \;\longrightarrow\; \text{code } [Z].$$

It holds for every algebraic cycle, with no restriction on k. The film's line "Geometry $\to$ code. Always. No exceptions." is this theorem.

In codimension one the same class has a second, more explicit name. If Z is a divisor, then

$$[Z] \;=\; c_1(\mathcal{O}_X(Z)),$$

the first Chern class of the associated line bundle. The exponential sequence produces every integral $(1,1)$ class in this way, and GAGA promotes the holomorphic bundle to an algebraic divisor. That is why $k=1$ is settled over $\mathbb{Z}$, not only over $\mathbb{Q}$. There is no analogous exponential sequence for $k \geq 2$.

What "decode" would mean

The cycle class map always runs in one direction. Decoding the final class would mean inverting it on the Hodge subspace: given

$$\gamma \in H^{2k}(X, \mathbb{Q}) \cap H^{k,k}(X),$$

produce finitely many subvarieties $Z_i \subset X$ of codimension k and rationals $a_i \in \mathbb{Q}$ such that

$$\gamma \;=\; \sum_i a_i [Z_i].$$

That is the Hodge conjecture, the Clay statement. The coefficients are rational. The last card of the video is exactly this request:

$$\text{Hodge class } \gamma \quad \longrightarrow \quad \text{algebraic cycle } Z.$$

The map cl is known. A section of cl over the Hodge classes is not known, except in the cases already listed in the film:

- $k = 1$, by Lefschetz $(1,1)$, integrally;
- $k = \dim X - 1$, by hard Lefschetz applied to the previous case;
- $\dim X \leq 3$, because those two degrees exhaust the interesting even cohomology.

In every remaining case — beginning with a codimension-2 class on a fourfold — the class $[Z]$ can be written down whenever Z is given, and cannot in general be reconstructed from γ alone. That is what "Decoding still required" refers to. It is not a missing formula for $[Z]$. It is the missing inverse over $\mathbb{Q}$.

One-line decode

The final subvariety class in the video is

$$[Z] \;=\; \mathrm{PD}\big([Z]_{\mathrm{hom}}\big) \;\in\; H^{2k}(X, \mathbb{Q}) \cap H^{k,k}(X),$$

the Poincaré dual of the fundamental class of an algebraic cycle. It is the source of every balanced signal that geometry can produce. The open problem is not to interpret that class. The open problem is to decide whether every balanced rational signal arises in this way.

The first open case

Let X be a general fourfold and let $\gamma \in \mathrm{Hdg}^2(X)$ be arbitrary. The missing object is a finite collection of surfaces $Z_i \subset X$ and rationals $a_i \in \mathbb{Q}$ such that

$$\gamma \;=\; \sum_i a_i [Z_i].$$

That equality is the Hodge conjecture in its first open case. Writing the Z_i down from γ would be a section of cl. The cycle class map itself is constructed by taking, for each closed irreducible subvariety $Z \subset X$ of codimension k, the fundamental class $[Z]_{\mathrm{hom}} \in H_{2n-2k}(X, \mathbb{Z})$, applying Poincaré duality to obtain $[Z] = \mathrm{PD}([Z]_{\mathrm{hom}})$, and extending $\mathbb{Q}$-linearly from irreducible subvarieties to $Z^k(X)_{\mathbb{Q}}$.

The integral conjecture is a different sentence

The integral Hodge conjecture asserts that

$$\mathrm{cl}_{\mathbb{Z}} \colon CH^k(X) \longrightarrow H^{2k}(X, \mathbb{Z})$$

is surjective onto $H^{2k}(X, \mathbb{Z}) \cap H^{k,k}(X)$. A class counts only if the coefficients are integers.

This sentence is not the Clay problem, and it is not open in general. Atiyah–Hirzebruch produced torsion counterexamples. Kollár produced non-torsion counterexamples on very general hypersurfaces in $\mathbb{P}^4$: integral Hodge classes which become algebraic only after multiplication by an integer greater than 1. If $m\gamma$ is algebraic for some $m \neq 0$, then γ is algebraic over $\mathbb{Q}$. Those examples do not disprove the rational conjecture.

On the three-chain sextic $X = V(F) \subset \mathbb{P}^5$, with plane $\Pi = V(x_3, x_4, x_5)$ and residual surface S satisfying $[S] = h^2 - [\Pi]$, the rational primitive projection

$$\alpha = [\Pi] - \tfrac{1}{6}h^2$$

lies in $H^4(X, \mathbb{Q})$, not a priori in $H^4(X, \mathbb{Z})$. The factor $\frac{1}{6}$ is legal for Clay and illegal as an integral class. Clearing it gives

$$\beta := 6\alpha = h^2 - 6[\Pi] = [S] - 5[\Pi].$$

Then $\langle \beta, h^2 \rangle = 0$ and β is an integral combination of two surfaces, so β lies in the image of $\mathrm{cl}_{\mathbb{Z}}$. Neither α nor β is a miss for either conjecture.

The lattice $\mathbb{Z}h^2 + \mathbb{Z}[\Pi]$ has intersection matrix $\left(\begin{smallmatrix} 6 & 1 \\ 1 & 21 \end{smallmatrix}\right)$ and discriminant 125. The self-intersection 21 is c_2 of the normal bundle of the plane. Discriminant 125 leaves open whether the lattice is saturated in $H^4(X, \mathbb{Z})$. An unsaturated class would be an integral question only: it would still lie in $\mathbb{Q}\langle h^2, [\Pi] \rangle$, hence would be algebraic over $\mathbb{Q}$. No such class is produced in this note.

A Verification certificate

The formal development was machine-checked as follows.

- Repository:
  https://github.com/BenFrohman/HODGE
- Verified commit:
  `6923bcaab2e99de7b9a186ec2f76eedd26916f37`
- Lean toolchain: `leanprover/lean4:v4.22.0`

- Mathlib: tag `v4.22.0`, commit (from `lake-manifest.json`)
  `79e94a093aff4a60fb1b1f92d9681e407124c2ca`
- Continuous integration run (build, axiom audit, `sorry`/`axiom` scan; passed):
  https://github.com/BenFrohman/HODGE/actions/runs/41574094131
- SHA-256 of the GitHub source tarball of the verified commit, downloaded twice with identical result:
  https://github.com/BenFrohman/HODGE/archive/6923bcaab2e99de7b9a186ec2f76eedd26916f37.tar.gz
  `1320d1417b0458fb52b38a38c0d7b81191a9a032d4c642f3fb3cd2394cfc91f0`

The tarball hash refers to the verified commit rather than to a release tag, since a document cannot contain the hash of an archive that includes itself. The Lean sources are unchanged between that commit and the release tag.

The development contains no `sorry` and no user-declared axioms. Every one of the 156 theorems and lemmas listed below depends only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound` (or on none). The table is the verbatim output of `lake env lean AxiomAudit.lean` (`#print axioms`) on the verified commit.

Lean declaration	`#print axioms`
`Hodge.CycleConstructor.construct`	`[propext, Classical.choice, Quot.sound]`
`Hodge.Datum.algebraicClasses_le_hodgeClasses`	`[propext, Classical.choice, Quot.sound]`
`Hodge.Datum.cl_isHodge`	`[propext, Classical.choice, Quot.sound]`
`Hodge.Datum.hodgeConjecture_codim_blind`	`[propext, Classical.choice, Quot.sound]`
`Hodge.Datum.hodgeConjecture_of_generators`	`[propext, Classical.choice, Quot.sound]`
`Hodge.constructor_of_hodgeConjecture`	`[propext, Classical.choice, Quot.sound]`
`Hodge.hodgeConjecture_of_constructor`	`[propext, Classical.choice, Quot.sound]`
`Hodge.Classical.P4Coord.ofNat_toCtorIdx`	`(no axioms)`
`Hodge.Classical.classical_islands`	`[propext, Classical.choice, Quot.sound]`
`Hodge.Classical.constructP4_coeff`	`[propext]`
`Hodge.Classical.constructP4_discharged`	`[propext, Classical.choice, Quot.sound]`
`Hodge.Classical.constructP4_section`	`[propext, Classical.choice, Quot.sound]`
`Hodge.Classical.constructProduct_section`	`[propext, Classical.choice, Quot.sound]`
`Hodge.Classical.construct_recovers`	`[propext, Classical.choice, Quot.sound]`
`Hodge.Classical.constructions_are_sections`	`[propext, Classical.choice, Quot.sound]`
`Hodge.Classical.expansion`	`[propext, Classical.choice, Quot.sound]`

Lean declaration	#print axioms
Hodge.Classical.instCycleConstructorProdRatKleinQuadric	[propext, Classical.choice, Quot.sound]
Hodge.Classical.instCycleConstructorProdRatProductOfPlanes	[propext, Classical.choice, Quot.sound]
Hodge.Classical.instCycleConstructorRatProjectiveFourSpace	[propext, Classical.choice, Quot.sound]
Hodge.Classical.kleinQuadric_hodgeConjecture	[propext, Classical.choice, Quot.sound]
Hodge.Classical.productOfPlanes_hodgeConjecture	[propext, Classical.choice, Quot.sound]
Hodge.Classical.projectiveFourSpace_hodgeConjecture	[propext, Classical.choice, Quot.sound]
Hodge.HodgeConjecture.general_fourfold_iff_clay	[propext, Classical.choice, Quot.sound]
Hodge.Classical.sections_recover	[propext, Classical.choice, Quot.sound]
Hodge.CycleSection.is_section	[propext, Classical.choice, Quot.sound]
Hodge.instCycleConstructorOfCycleSection	[propext, Classical.choice, Quot.sound]
Hodge.DerivedSurfaces.every_listed_host_is_a_fourfold	(no axioms)
Hodge.DerivedSurfaces.every_listed_object_is_a_surface	(no axioms)
Hodge.DerivedSurfaces.inventory_card	(no axioms)
Hodge.DerivedSurfaces.residual_host_is_fourfold	(no axioms)
Hodge.DerivedSurfaces.residual_is_surface	(no axioms)
Hodge.DerivedSurfaces.residual_not_partner	(no axioms)
Hodge.Examples.instCycleConstructorRatTrivialObstruction	[propext, Classical.choice, Quot.sound]
Hodge.Examples.not_every_codim_ge_two	[propext, Classical.choice, Quot.sound]
Hodge.Examples.trivialObstruction_hodge	[propext, Classical.choice, Quot.sound]
Hodge.Examples.zeroCycle_is_not_a_variety_sticker	(no axioms)
Hodge.Examples.zeroCycle_not_hodge	[propext, Classical.choice, Quot.sound]
Hodge.FermatQuintic.F5_eq_G	(no axioms)
Hodge.FermatQuintic.G_not_euler_degree5	[propext, Quot.sound]
Hodge.FermatQuintic.X_5_name_ne_T_F	(no axioms)
Hodge.FermatQuintic.affine_cone_isolated_at_origin	[propext, Classical.choice, Quot.sound]
Hodge.FermatQuintic.b4_numeral	(no axioms)
Hodge.FermatQuintic.canonical_degree	(no axioms)
Hodge.FermatQuintic.euler_homogeneous	[propext]
Hodge.FermatQuintic.h22_full_numeral	(no axioms)
Hodge.FermatQuintic.h31_numeral	(no axioms)
Hodge.FermatQuintic.h40_numeral	(no axioms)
Hodge.FermatQuintic.jacobian_certificate	[propext, Classical.choice, Quot.sound]
Hodge.FermatQuintic.this_is_not_general_fourfold	(no axioms)
Hodge.Fermat.Coord.ofNat_toCtorIdx	(no axioms)
Hodge.Fermat.Z1_pairing	[propext]
Hodge.Fermat.Z2_pairing	[propext]

Lean declaration	#print axioms
Hodge.Fermat.construct_section	[propext, Classical.choice, Quot.sound]
Hodge.Fermat.contains_two_planes	[propext, Classical.choice, Quot.sound]
Hodge.Fermat.fermatQuartic_hodge	[propext, Classical.choice, Quot.sound]
Hodge.Fermat.twoPlanes_hodge	[propext, Classical.choice, Quot.sound]
Hodge.HodgeConjecture.LefschetzTwoTwo_iff	[propext, Classical.choice, Quot.sound]
Hodge.HodgeConjecture.classical_fourfolds	[propext, Classical.choice, Quot.sound]
Hodge.HodgeConjecture.general_fourfold_iff	[propext, Classical.choice, Quot.sound]
Hodge.HodgeConjecture.klein	[propext, Classical.choice, Quot.sound]
Hodge.HodgeConjecture.of_constructor	[propext, Classical.choice, Quot.sound]
Hodge.HodgeConjecture.of_section	[propext, Classical.choice, Quot.sound]
Hodge.HodgeConjecture.p4	[propext, Classical.choice, Quot.sound]
Hodge.HodgeConjecture.product	[propext, Classical.choice, Quot.sound]
Hodge.LefschetzOneOne.construct	[propext, Classical.choice, Quot.sound]
Hodge.LefschetzOneOne.is_codim_one	[propext, Classical.choice, Quot.sound]
Hodge.hodgeConjecture_codim_one	[propext, Classical.choice, Quot.sound]
Hodge.instCycleConstructorOfLefschetzOneOne	[propext, Classical.choice, Quot.sound]
Hodge.instLefschetzOneOneRatK3Surface	[propext, Classical.choice, Quot.sound]
Hodge.k3Surface_codim	[propext, Classical.choice, Quot.sound]
Hodge.k3Surface_hodge	[propext, Classical.choice, Quot.sound]
Hodge.Geometry.Variety.hodgeConjecture_iff	[propext, Classical.choice, Quot.sound]
Hodge.Geometry.cycleClass_is_hodge	[propext, Classical.choice, Quot.sound]
Hodge.Geometry.every_datum_wraps	[propext, Classical.choice, Quot.sound]
Hodge.Geometry.kleinQuadric_hodge	[propext, Classical.choice, Quot.sound]
Hodge.Geometry.onVarieties_false_on_Rat	[propext, Classical.choice, Quot.sound]
Hodge.Geometry.productOfPlanes_hodge	[propext, Classical.choice, Quot.sound]
Hodge.Geometry.projectiveFourSpace_hodge	[propext, Classical.choice, Quot.sound]
Hodge.Geometry.wrap_data	[propext, Classical.choice, Quot.sound]

Lean declaration	#print axioms
Hodge.Geometry.wrapped	[propext, Classical.choice, Quot.sound]
Hodge.Geometry.zeroGadget_not_hodge	[propext, Classical.choice, Quot.sound]
Hodge.Hassett.P5Coord.ofNat_toCtorIdx	(no axioms)
Hodge.Hassett.construct_section	[propext, Classical.choice, Quot.sound]
Hodge.Hassett.planeSpan_hodge	[propext, Classical.choice, Quot.sound]
Hodge.SpecialSextic.head_plane_not_on_host	[propext, Classical.choice, Quot.sound]
Hodge.SpecialSextic.head_plane_point	[propext, Classical.choice, Quot.sound]
Hodge.HostSplit.Z_uncomputed	(no axioms)
Hodge.HostSplit.containers_differ	(no axioms)
Hodge.HostSplit.quintic_b4	(no axioms)
Hodge.HostSplit.quintic_h22	(no axioms)
Hodge.HostSplit.quintic_h31	(no axioms)
Hodge.HostSplit.sextic_b4	(no axioms)
Hodge.HostSplit.sextic_h22	(no axioms)
Hodge.HostSplit.sextic_length	(no axioms)
Hodge.HostSplit.sextic_r12	(no axioms)
Hodge.HostSplit.sextic_r18	(no axioms)
Hodge.HostSplit.sextic_r6	(no axioms)
Hodge.Datum.hodgeConjecture_of_integral_generation	[propext, Classical.choice, Quot.sound]
Hodge.Integral.integral_unsolvable	[propext]
Hodge.Integral.rational_does_not_imply_integral	[propext, Classical.choice, Quot.sound]
Hodge.Integral.rational_solvable	[propext, Classical.choice, Quot.sound]
Hodge.Klein.Coord.ofNat_toCtorIdx	(no axioms)
Hodge.Klein.cycleOf_coeff	[propext]
Hodge.Klein.cycleOf_construct	[propext]
Hodge.Klein.ideals_cover	[propext]
Hodge.Klein.ideals_disjoint	[propext]
Hodge.Klein.pi'_on_quadric	[propext, Classical.choice, Quot.sound]
Hodge.Klein.pi_on_quadric	[propext, Classical.choice, Quot.sound]
Hodge.DimLeThree.construct	[propext, Classical.choice, Quot.sound]
Hodge.HardLefschetzComplement.construct	[propext, Classical.choice, Quot.sound]
Hodge.hodgeConjecture_dim_le_three	[propext, Classical.choice, Quot.sound]
Hodge.hodgeConjecture_hardLefschetz	[propext, Classical.choice, Quot.sound]
Hodge.instCycleConstructorOfDimLeThree	[propext, Classical.choice, Quot.sound]
Hodge.instCycleConstructorOfHardLefschetzComplement	[propext, Classical.choice, Quot.sound]
Hodge.HodgeConjecture.frohman_two_two_spans	[propext, Classical.choice, Quot.sound]

Lean declaration	#print axioms
Hodge.HodgeConjecture.frohman_two_two_spans_iff	[propext, Classical.choice, Quot.sound]
Hodge.HodgeConjecture.named_fourfolds	[propext, Classical.choice, Quot.sound]
Hodge.SexticModuli.dim_PGL6	(no axioms)
Hodge.SexticModuli.eq_15625_of_eq_five_pow_six	(no axioms)
Hodge.SexticModuli.equation_space_dim	(no axioms)
Hodge.SexticModuli.fixed_plane_naive_dim	(no axioms)
Hodge.SexticModuli.milnor_number	(no axioms)
Hodge.SexticModuli.naive_git_dim	(no axioms)
Hodge.SexticModuli.plane_restriction_monomials	(no axioms)
Hodge.SexticModuli.sextic_monomials	(no axioms)
Hodge.SpecialSextic.dF_dx0_formula	(no axioms)
Hodge.SpecialSextic.dF_dx1_formula	(no axioms)
Hodge.SpecialSextic.dF_dx2_formula	(no axioms)
Hodge.SpecialSextic.dF_dx3_formula	(no axioms)
Hodge.SpecialSextic.dF_dx4_formula	(no axioms)
Hodge.SpecialSextic.dF_dx5_formula	(no axioms)
Hodge.SpecialSextic.euler_homogeneous	[propext]
Hodge.SpecialSextic.jacobian_lock_is_not_general_fourfold	(no axioms)
Hodge.SpecialSextic.jacobian_vanishes_only_at_origin	[propext, Classical.choice, Quot.sound]
Hodge.SpecialSexticMembership.F_factors	[propext]
Hodge.SpecialSexticMembership.F_factors_minus1	[propext, Quot.sound]
Hodge.SpecialSexticMembership.F_mem_plane	[propext, Quot.sound]
Hodge.SpecialSexticMembership.F_mem_plane_minus1	[propext, Quot.sound]
Hodge.SpecialSexticMembership.contains_two_planes	[propext, Quot.sound]
Hodge.SpecialSextic.F_factors	[propext]
Hodge.SpecialSextic.F_factors_minus1	[propext, Quot.sound]
Hodge.SpecialSextic.F_mem_plane	[propext, Quot.sound]
Hodge.SpecialSextic.F_mem_plane_minus1	[propext, Quot.sound]
Hodge.SpecialSextic.affine_cone_isolated_at_origin	[propext, Classical.choice, Quot.sound]
Hodge.SpecialSextic.constructThree_section	[propext, Classical.choice, Quot.sound]
Hodge.SpecialSextic.construct_section	[propext, Classical.choice, Quot.sound]
Hodge.SpecialSextic.contains_two_planes	[propext, Quot.sound]
Hodge.SpecialSextic.five_pow_six	[propext]
Hodge.SpecialSextic.gradPair_origin	[propext, Classical.choice, Quot.sound]
Hodge.SpecialSextic.gradient_only_origin	[propext, Classical.choice, Quot.sound]
Hodge.SpecialSextic.gram_det_numeral	(no axioms)
Hodge.SpecialSextic.hypersurface_contains_the_plane	[propext, Quot.sound]
Hodge.SpecialSextic.hypersurface_contains_the_sign_plane	[propext, Quot.sound]
Hodge.SpecialSextic.planeSpan_hodge	[propext, Classical.choice, Quot.sound]
Hodge.SpecialSextic.special_NL_sextic_coefficient_shadow	[propext, Classical.choice, Quot.sound]

Lean declaration	`#print axioms`
`Hodge.SpecialSextic.threeSpan_hodge`	`[propext, Classical.choice, Quot.sound]`